

COMP1831 – Technologies for Anti-Money Laundering and Financial Crime

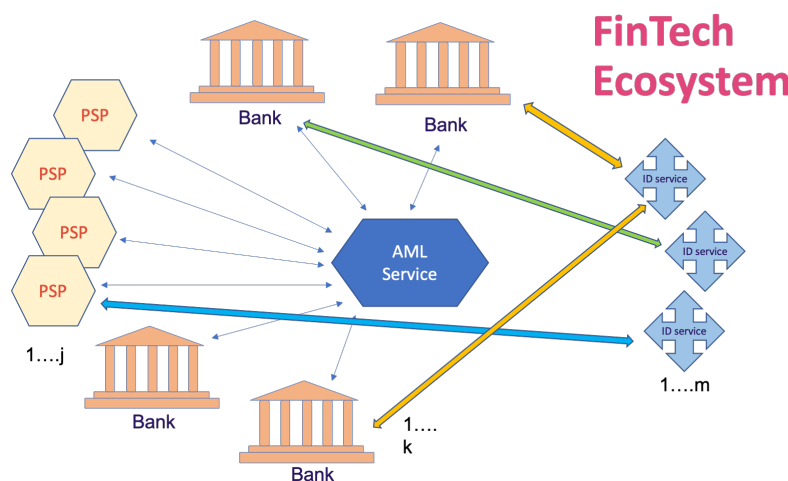
Assessment Specification 2025-26

Assessment for the module consists of a **single submission (100% of the mark)**. This comprises 2 components as below:

1. An individual Report (approx. 2.5-3 thousand words) that will count for **70% of your final mark**.
2. Your Lab Portfolio (consists of your lab exercise implementation), comprising working versions of all the exercises you will execute during the term. That component will count for **30% of your final mark**.

Detailed Specification of REPORT (70% of total mark)

Consider the FinTech ecosystem broadly described in the Figure below. This represents a network of financial institutions and service providers that operate in the UK (the ecosystem may include more types of financial intermediaries, however the 3 provided here (bank, payments service provider, ID verification service provider) are the minimum). We assume **k** banks, **j** Payment Service Providers and **m** ID verification service providers.



These entities serve multiple commercial and private customers of different types (businesses, other intermediaries, individuals etc), in effect offering financial services to the market. (These customers are not represented in the diagram for simplicity).

You are a FinTech start-up aiming to launch as an **“Anti-money laundering as a service”** provider, and are planning to file for an FCA license in the next few months. To that end, you are preparing a Report for the regulator (the UK Payments Systems Regulator (PSR)) where you will be outlining the case for why this license should be provided to you. The report should include (not necessarily in the order below):

1. A background section where you will outline the problem you are proposing to address; this should include brief information (e.g. based on available statistics) on the state of Money Laundering in the UK and international markets, making also reference to evidence on the reduced effectiveness of existing AML solutions (again using available evidence).
2. A high-level explanation of what your service / solution will do, who it is for, how it will be used and what technologies will it use (e.g. machine learning; differential privacy; blockchain; visual analytics; graph databases; federated learning; etc etc)
3. An outline of your data access needs. That will address the type of data you will be asking the regulator to allow you to use in order to deliver the service.
4. An outline of your approach to data privacy: this should address how (in terms of both technology and policy / governance) you will ensure the security and integrity of any sensitive data you will make use of (see 2 above)
5. A high-level architecture plan (e.g. business process flow diagram / infrastructure block diagram / data architecture diagram) demonstrating how the solution will work.
6. A more detailed elaboration of the tools and methodologies your solution will use (including for data preparation) to deliver what you have discussed in (2) above. Where possible, these tools and methodologies should be mapped to respective stages of the Money Laundering lifecycle.

Please note there will be no specific assumptions about details related to headcount (nr. of people you employ) or any limitations in costs associated with your solution. However, you should adopt reasonable estimates.

Outline Specification of LAB PORTFOLIO (30% of total mark)

Your Tutorial Portfolio will consist of exercises, the specification for which will be individually provided during the term. In these, you will be required to:

1. Deploy a specific piece of software and carry out a predefined set of steps (including, where applicable, installation of supporting libraries and setting up the working environment).
2. Execute an exercise following the instructions you will be given during the Lab Sessions

All exercises should be appended in a **single DOCX or PDF file, together with the REPORT.**

In each exercise, you would be typically required to submit:

- (i) screenshots of your working application;
- (ii) paste of code where appropriate AND links to working versions of your code.
- (iii) Text explanations of the steps followed

The full specifications of each exercise, together with the marks awarded for each will be provided during the term and will be discussed during Lab Sessions.

The exact number of exercises you need to include in your submission, as well as information on which ones will be shared with you 1 week prior to the deadline. It is therefore strongly advised that ALL exercises are completed during the term.

Deliverables.

REPORT (70%): A word-processed, report-style, **fully referenced (with in-text citation)** paper of 3500 words (+/- 10%). This excludes Appendices and References.

LAB (30%): A word-processed document outlining lab execution steps with brief explanations, screenshots, pasted code (where appropriate) and links to working versions of your code. (This part has no word limit).

The two components above should be submitted as a SINGLE FILE (docx or pdf).

Grading Criteria:

70%+. An excellent attempt demonstrating a clear understanding of the requirements of the assignment. An impressive demonstration of research, organisation, initiative and application.

60-69%. A good attempt demonstrating a strong understanding of the requirements of the assignment. A good display of research, organisation, initiative and application.

50-59%. A satisfactory attempt demonstrating an effective understanding of the assignment. An acceptable display of research, organisation, initiative and application.

40-49%. A basic understanding of the assignment and its effect. Research, analysis and initiative limited.

30-39%. Poor understanding of the requirements of the assignment. Possibly some confusion and much irrelevant material. A poor display of research, organisation, initiative and application.

0-29%. Little to no understanding of the requirements of the assignment. Little effort involved in research, organisation, initiative or application.